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Abstract: In order to improve the detection accuracy of the network, it proposes multi-scale feature fusion and attention mechanism net (MFANet) based on deep learning, which integrates pyramid module and channel attention mechanism effectively. Pyramid module is designed for feature fusion in the channel and space dimensions. Channel attention mechanism obtains feature maps in different receptive fields, which divides each feature map into two groups and uses different convolutions to obtain weights. Experimental results show that our strategy boosts state-of-the-arts by 1–2% box AP on object detection benchmarks. Among them, the accuracy of MFANet reaches 34.2% in box AP on COCO dataset. Compared with the current typical algorithms, the proposed method achieves significant performance in detection accuracy.
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